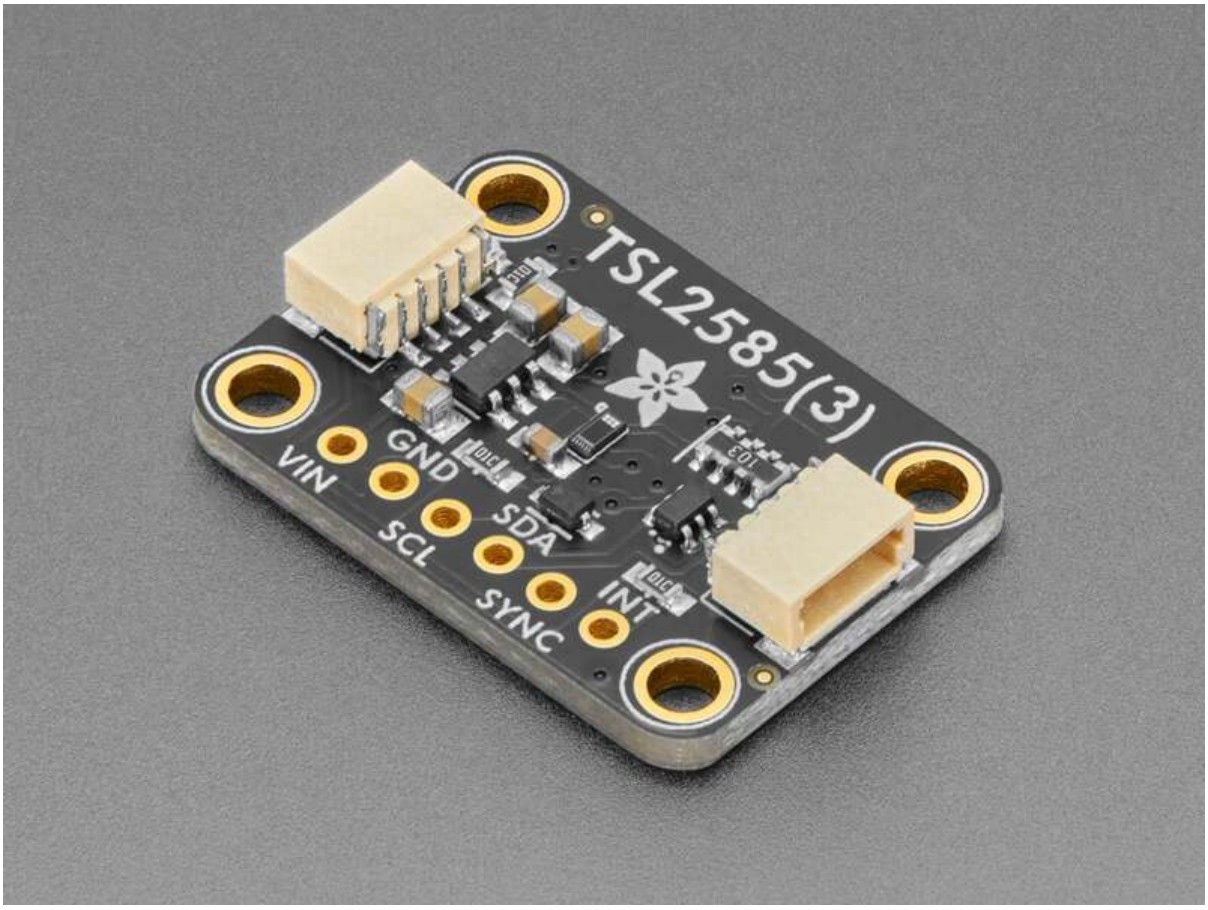
Adafruit TSL2585 Digital UVA and Ambient Light Sensor

Created by Liz Clark



https://learn.adafruit.com/adafruit-tsl2585-digital-uva-and-ambient-light-sensor

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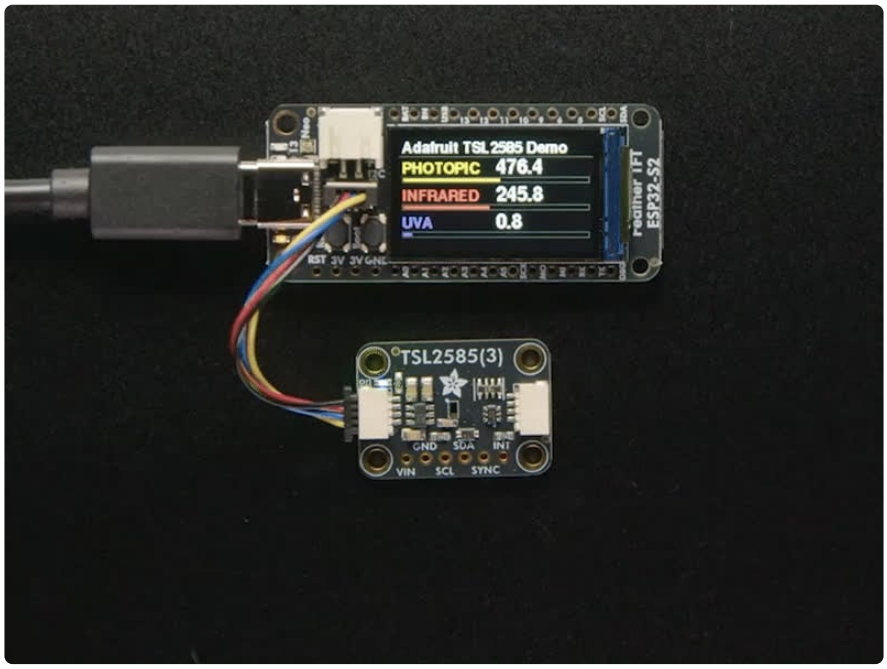
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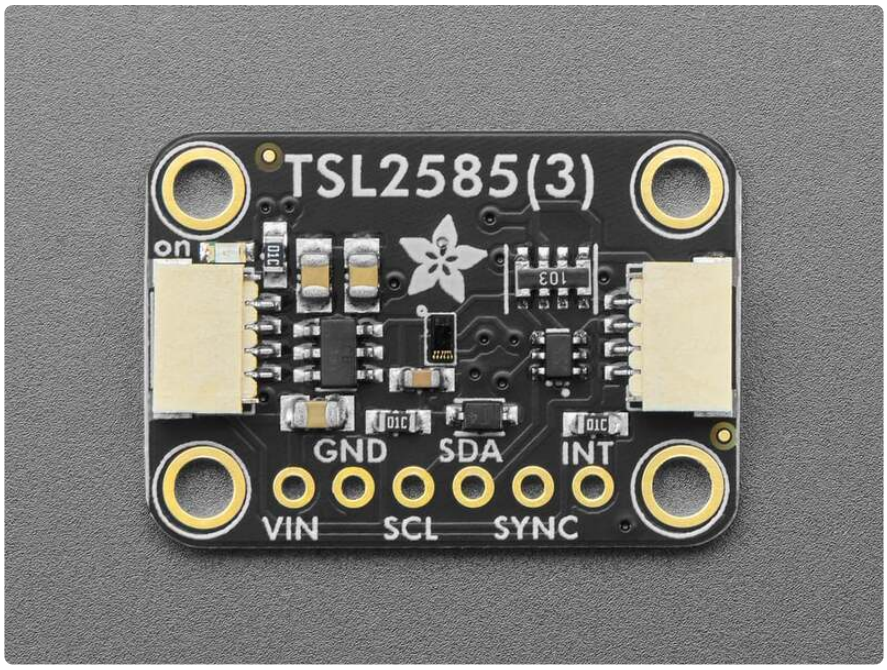
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Overview

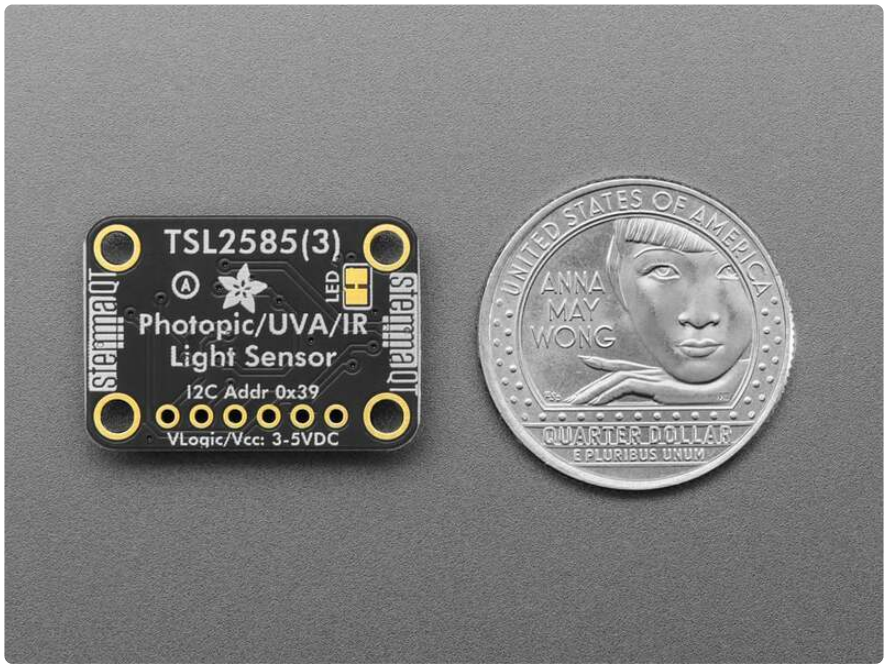


The **Adafruit TSL2585 Digital UVA and Ambient Light Sensor** is a great "three-in-one" light sensor that has ambient light sensing, infrared channel and UVA (315nm-400nm). In particular it's not that common to get UVA narrow-band sensing on an I2C sensor, so this is great for outdoor light monitoring. It can also be used to measure curing lights, blacklights and reptile lamps!

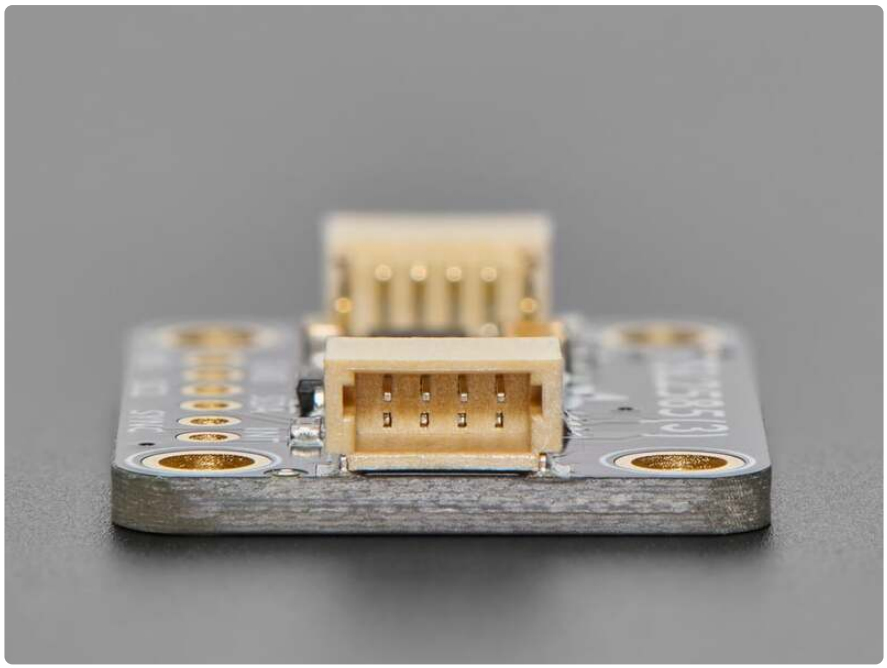


Please Note: this sensor will give 'counts' for the ambient light sensor, and is not lux calibrated, so you can use it for measurements but it wont convert to lux. The UVA sensor does have a calibration byte, but only for 'factory corrected' counts, not µW/cm^2.

The sensor has an auto-gain system with up to x4096 gain, plus adjustable integration time, so it can detect down to 1 millilux.

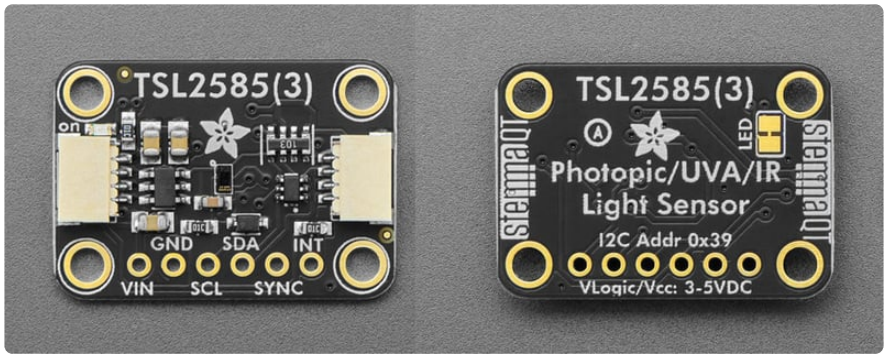


This chip is tiny and normally uses 1.8V power and logic so we've wrapped it into a tasty Stemma QT breakout board with SparkFun qwiic () compatible STEMMA QT **()** connectors for the I2C bus **so you don't even need to solder.** Just wire up to your favorite micro with a plug-and-play cable to get data ASAP. For a no-solder experience, just wire up to your favorite micro using a STEMMA QT adapter cable (). The Stemma QT connectors also mean the TSL2585 can be used with our various associated accessories (). QT Cable is not included, but we have a variety in the shop ().



Communicating to the sensor is done over I2C, [check out our Arduino and CircuitPython/Python code ()](#) to get started fast and read data over I2C in a jiffy.

Pinouts



The default I2C address is **0x39**.

Power Pins

- **VIN** - this is the power pin. To power the board, give it the same power as the logic level of your microcontroller - e.g. for a 5V microcontroller like Arduino, use 5V. For a 3.3V microcontroller, use 3.3V.
- **GND** - common ground for power and logic.

I2C Logic Pins

- **SCL** - I2C clock pin, connect to your microcontroller I2C clock line. This pin is level shifted so you can use 3-5V logic, and there's a **10K pullup** on this pin.
- **SDA** - I2C data pin, connect to your microcontroller I2C data line. This pin is level shifted so you can use 3-5V logic, and there's a **10K pullup** on this pin.
- STEMMA QT () - These connectors allow you to connect to dev boards with **STEMMA QT** connectors or to other things with various associated accessories ().

Other Pins

- **SYNC** - The **SYNC** pin is an input that can trigger synchronous readings from the TSL2585 with an external pulse. It can also be configured as an open drain GPIO pin.
- **INT** - The interrupt pin is an open drain output and can be configured to alert you when a sensor conversion is complete.

Power LED and Jumper

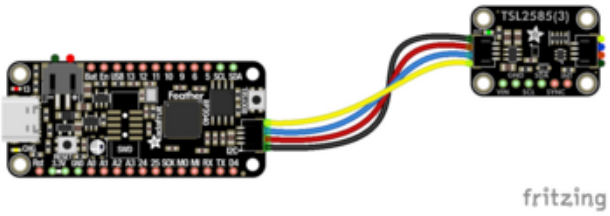
- **Power LED -** In the upper left corner, above the STEMMA QT connector, on the front of the board, is the power LED, labeled **on**. It is a green LED.
- **LED jumper** - This jumper is located on the back of the board and is labeled **LED** on the board silk. Cut the trace on this jumper to cut power to the "**on**" LED.

CircuitPython and Python

It's easy to use the **TSL2585** with Python or CircuitPython, and the Adafruit_CircuitPython_TSL2585 () module. This module allows you to easily write Python code that allows you to read the TSL2585 UVA sensor. You can use this sensor with any CircuitPython microcontroller board or with a computer that has GPIO and Python thanks to Adafruit_Blinka, our CircuitPython-for-Python compatibility library ().

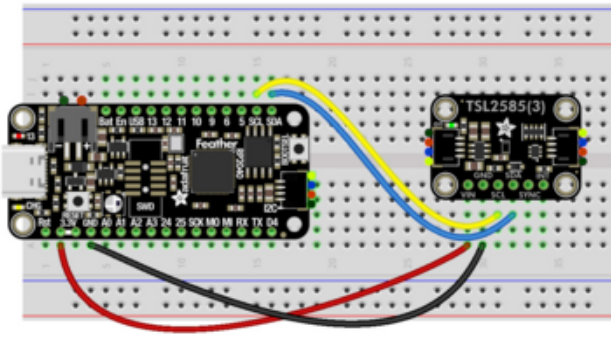
CircuitPython Microcontroller Wiring

First, wire up an TSL2585 to your board exactly as shown below. Here's an example of wiring a Feather RP2040 to the TSL2585 with I2C using one of the handy **STEMMA QT** () connectors:



Board STEMMA 3V to **breakout STEMMA VIN (red wire)**
Board STEMMA GND to **breakout STEMMA GND (black wire)**
Board STEMMA SCL to **breakout STEMMA SCL (yellow wire)**
Board STEMMA SDA to **breakout STEMMA SDA (blue wire)**

You can also use standard **0.100" pitch** headers to wire it up on a breadboard:

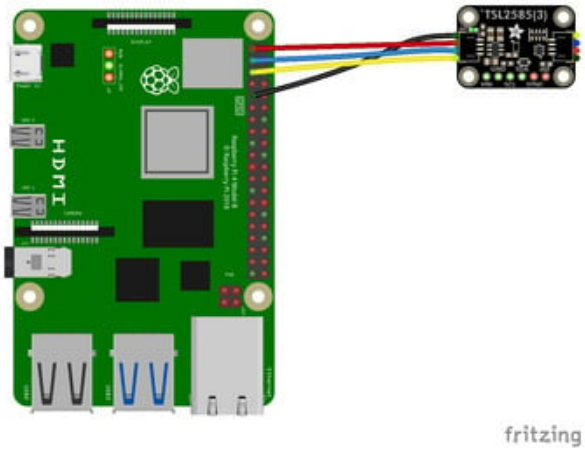


Board 3V to **breakout VIN (red wire)**
Board GND to **breakout GND (black wire)**
Board SCL to **breakout SCL (yellow wire)**
Board SDA to **breakout SDA (blue wire)**

Python Computer Wiring

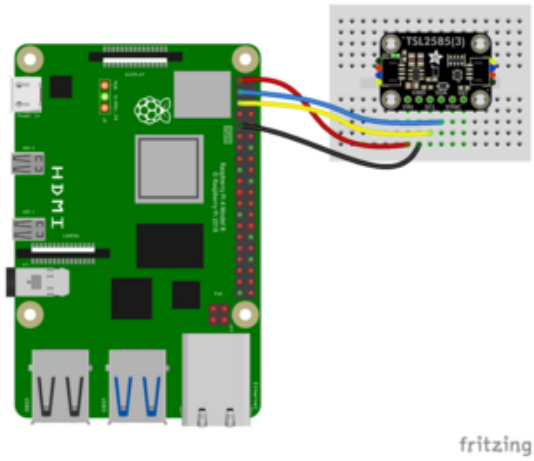
Since there's dozens of Linux computers/boards you can use, below shows wiring for Raspberry Pi. For other platforms, [please visit the guide for CircuitPython on Linux to see whether your platform is supported]() ().

Here's the Raspberry Pi wired to the sensor using I2C and a **STEMMA QT** () connector:



Pi 3V to **breakout STEMMA VIN (red wire)**
Pi GND to **breakout STEMMA GND (black wire)**
Pi SCL to **breakout STEMMA SCL (yellow wire)**
Pi SDA to **breakout STEMMA SDA (blue wire)**

Finally, here is an example of how to wire up a Raspberry Pi to the sensor using a solderless breadboard:



Pi 3V to **breakout VIN (red wire)**
Pi GND to **breakout GND (black wire)**
Pi SCL to **breakout SCL (yellow wire)**
Pi SDA to **breakout SDA (blue wire)**

Python Installation of TSL2585 Library

You'll need to install the **Adafruit_Blinka** library that provides the CircuitPython support in Python. This may also require enabling I2C on your platform and verifying you are running Python 3. Since each platform is a little different, and Linux changes often, please visit the CircuitPython on Linux guide to get your computer ready ()!

Once that's done, from your command line run the following command:

- `pip3 install adafruit-circuitpython-tsl2585`

If your default Python is version 3, you may need to run `pip` instead. Make sure you aren't trying to use CircuitPython on Python 2.x, it isn't supported!

CircuitPython Usage

To use with CircuitPython, you need to first install the TSL2585 library and its dependencies into the **lib** folder on your **CIRCUITPY** drive. Then you need to update **code.py** with the example script.

Thankfully, we can do this in one go. In the example below, click the **Download Project Bundle** button below to download the necessary libraries and the **code.py** file in a zip file. Extract the contents of the zip file, and copy the **entire lib folder** and the **code.py** file to your **CIRCUITPY** drive.

Your **CIRCUITPY/lib** folder should contain the following folders and file:

- **adafruit_bus_device/**
- **adafruit_register/**
- **adafruit_tsl2585.mpy**



Python Usage

Once you have the library `pip3` installed on your computer, copy or download the following example to your computer, and run the following, replacing **code.py** with whatever you named the file:

```
python3 code.py
```

Example Code

If running CircuitPython: Once everything is saved to the **CIRCUITPY** drive, connect to the serial console () to see the data printed out!

If running Python: The console output will appear wherever you are running Python.

```
# SPDX-FileCopyrightText: Copyright (c) 2026 Liz Clark for Adafruit Industries
#
# SPDX-License-Identifier: MIT

"""Print three channel measurements from the TSL2585."""

import time

import board

import adafruit_tsl2585

i2c = board.I2C()
sensor = adafruit_tsl2585.TSL2585(i2c)

print(f"Found TSL2585, revision {sensor.revision_id}")

while True:
    if sensor.data_ready:
        data = sensor.measurement
        print(f"Photopic: {data.photopic}, IR: {data.infrared}, UVA: {data.uva}")
        if data.photopic_saturated or data.infrared_saturated or data.uva_saturated:
            print("Warning: at least one channel saturated")
        print()
    time.sleep(0.5)
```

https://github.com/adafruit/Adafruit_CircuitPython_TSL2585/blob/main/examples/tsl2585_simpletest.py

In the example, the sensor is instantiated on I2C. Then, in the loop, the photopic, IR and UVA readings are printed to the serial console.

Text emphasized with a blue info:

Please Note: this sensor will give 'counts' for the ambient light sensor, and is not lux calibrated, so you can use it for measurements but it won't convert to Lux. The UVA sensor does have a calibration byte, but only for 'factory corrected' counts, not µW/cm^2.

```
Auto-reload is on. Simply save files over USB to run them or enter REPL to disable.
code.py output:
Found TSL2585, revision 17
Photopic 16461, IR 13994, UVA 15228

Photopic 17352, IR 16549, UVA 20161

Photopic 15792, IR 14876, UVA 19147

Photopic 20031, IR 20747, UVA 14179

Photopic 22517, IR 23230, UVA 16534

Photopic 23688, IR 24736, UVA 17911

Photopic 25424, IR 12940, UVA 19754

Photopic 14607, IR 13076, UVA 22014

Photopic 16850, IR 15121, UVA 12628

Photopic 17184, IR 15412, UVA 12853

Photopic 16437, IR 15747, UVA 12585

Photopic 15353, IR 14846, UVA 23490
```

Python Docs

Python Docs ()

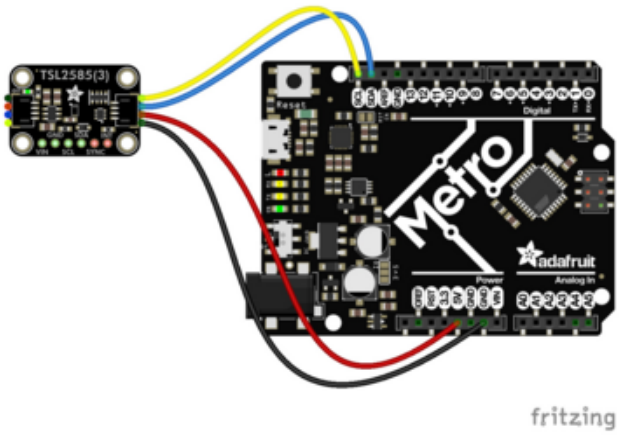
Arduino

Using the TSL2585 UVA and ambient light sensor with Arduino involves wiring up the sensor to your Arduino-compatible microcontroller, installing the Adafruit_TSL2585 () l ibrary and running the provided example code.

Wiring

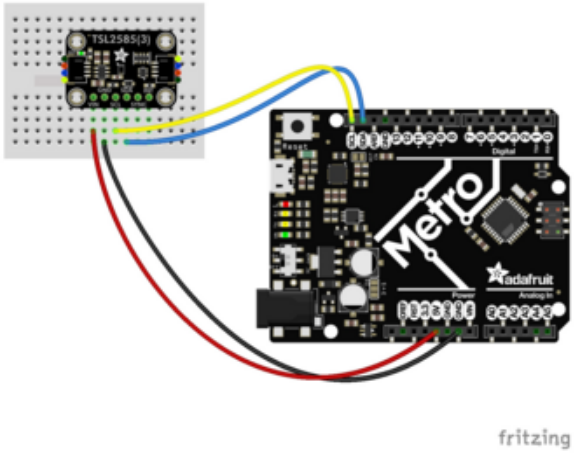
Wire as shown for a **5V** board like an Uno. If you are using a **3V** board, like an Adafruit Feather, wire the board's 3V pin to the TSL2585 VIN.

Here is an Adafruit Metro wired up to the TSL2585 using the STEMMA QT connector:



Board 5V to **breakout STEMMA VIN (red wire)**
Board GND to **breakout STEMMA GND (black wire)**
Board SCL to **breakout STEMMA SCL (yellow wire)**
Board SDA to **breakout STEMMA SDA (blue wire)**

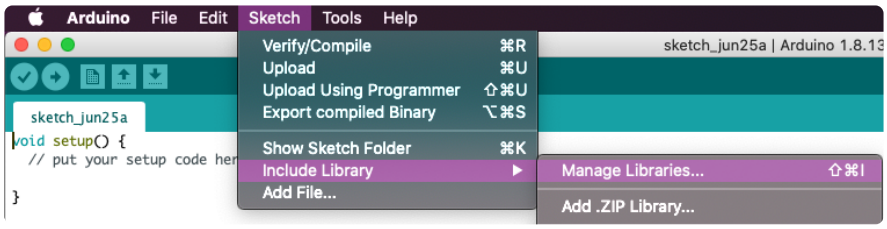
Here is an Adafruit Metro wired up using a solderless breadboard:



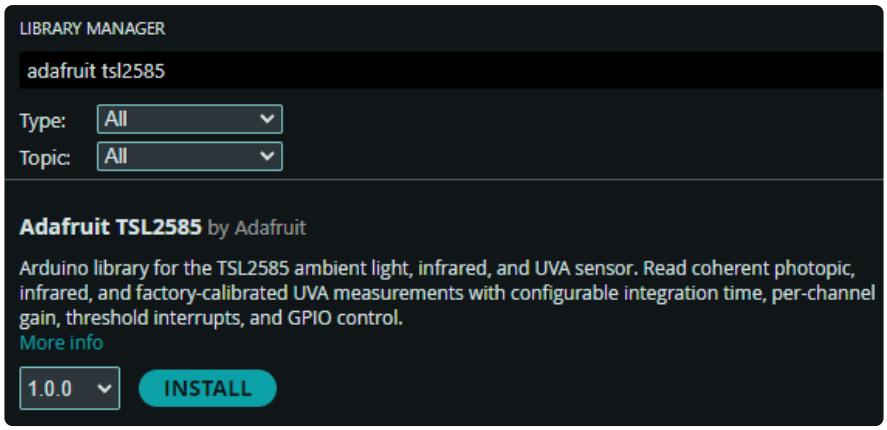
Board 5V to **breakout VIN (red wire)**
Board GND to **breakout GND (black wire)**
Board SCL to **breakout SCL (yellow wire)**
Board SDA to **breakout SDA (blue wire)**

Library Installation

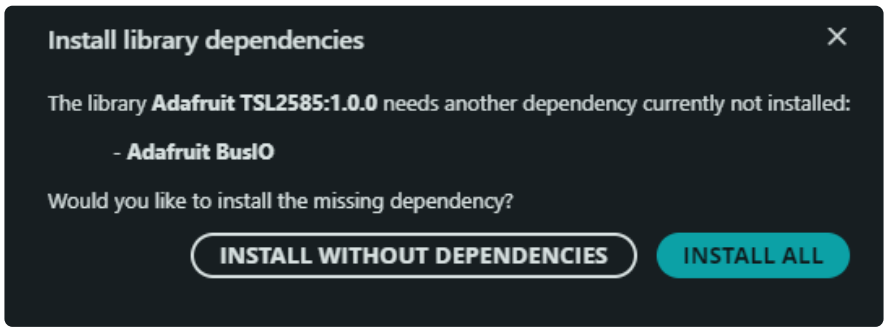
You can install the **Adafruit TSL2585** library for Arduino using the Library Manager in the Arduino IDE.



Click the **Manage Libraries ...** menu item, search for **Adafruit TSL2585**, and select the **Adafruit TSL2585** library:



If asked about dependencies, click "**Install all**".



If the "Dependencies" window does not come up, then you already have the dependencies installed.

Text emphasized with a yellow exclamation:

If the dependencies are already installed, you must make sure you update them through the Arduino Library Manager before loading the example!

Example Code

```
#include <Adafruit_TSL2585.h>

Adafruit_TSL2585 tsl2585;

void setup() {
  Serial.begin(115200);
  // Wait for the Serial Monitor to open on native USB boards.
  // Remove this while (!Serial) loop to run without a USB connection.
```

```
  while (!Serial) {
    delay(10);
  }
  delay(250);

  Serial.println("Adafruit TSL2585 photopic, infrared, and UVA sensor test");

  if (!tsl2585.begin()) {
    Serial.println("Could not find a TSL2585. Check the wiring and I2C address.");
    while (true) {
      delay(10);
    }
  }

  Serial.print("Device ID: 0x");
  Serial.println(tsl2585.getDeviceID(), HEX);
  Serial.print("Revision ID: 0x");
  Serial.println(tsl2585.getRevisionID(), HEX);
  Serial.print("Factory UVA calibration byte: ");
  Serial.println(tsl2585.getUVCalibration());

  // Register-based results support integration times from 0.25 ms to 90 ms.
  if (!tsl2585.setIntegrationTime(50)) {
    Serial.println("Could not set the integration time.");
    while (true) {
      delay(10);
    }
  }
  Serial.print("Integration time: ");
  Serial.print(tsl2585.getIntegrationTime(), 2);
  Serial.println(" ms");

  Serial.println();
  Serial.println("Photopic 1x\tInfrared 1x\tUVA 1x\tSaturated");
}

void loop() {
  if (!tsl2585.dataReady()) {
    delay(100);
    return;
  }

  tsl2585_data_t data;
  if (tsl2585.readData(&data)) {
    bool saturated = false;
    if (data.photopic_saturated || data.infrared_saturated ||
        data.uva_saturated) {
      saturated = true;
    }

    Serial.print(data.photopic_1x, 1);
    Serial.print('\t');
    Serial.print(data.infrared_1x, 1);
    Serial.print('\t');
    Serial.print(data.uva_1x, 1);
    Serial.print('\t');
```

```
    if (saturated) {
      Serial.print("saturated");
    }
    Serial.println();
  }

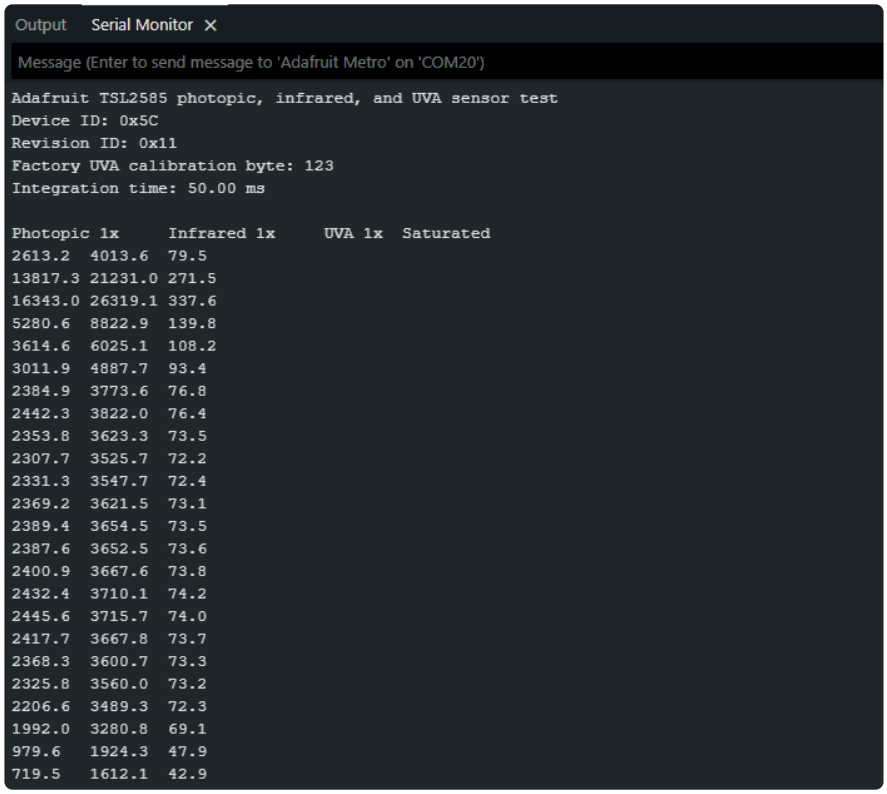
  delay(100);
}
```

https://github.com/adafruit/Adafruit_TSL2585/blob/main/examples/tsl2585_basic/tsl2585_basic.ino

Upload the sketch to your board and open up the Serial Monitor (**Tools -> Serial Monitor**) at 115200 baud. You should see that the sketch has found your connected TSL2585 sensor. Then, when data is ready, you'll see the phototopic, infrared and UVA counts printed to the Serial Monitor, as well as if the sensor is saturated or not.

Text emphasized with a blue info:

Please Note: this sensor will give 'counts' for the ambient light sensor, and is not lux calibrated, so you can use it for measurements but it won't convert to Lux. The UVA sensor does have a calibration byte, but only for 'factory corrected' counts, not µW/cm^2.



Arduino Docs

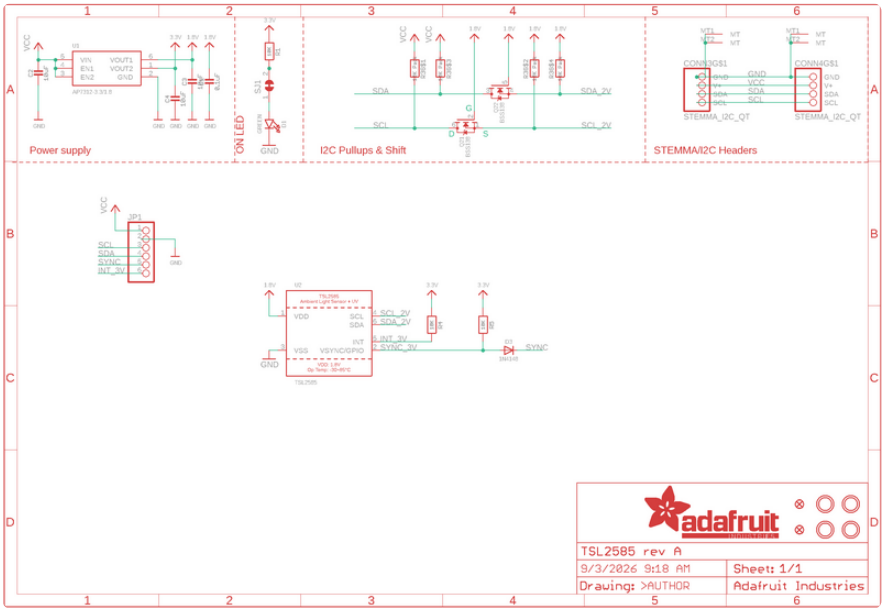
Arduino Docs ()

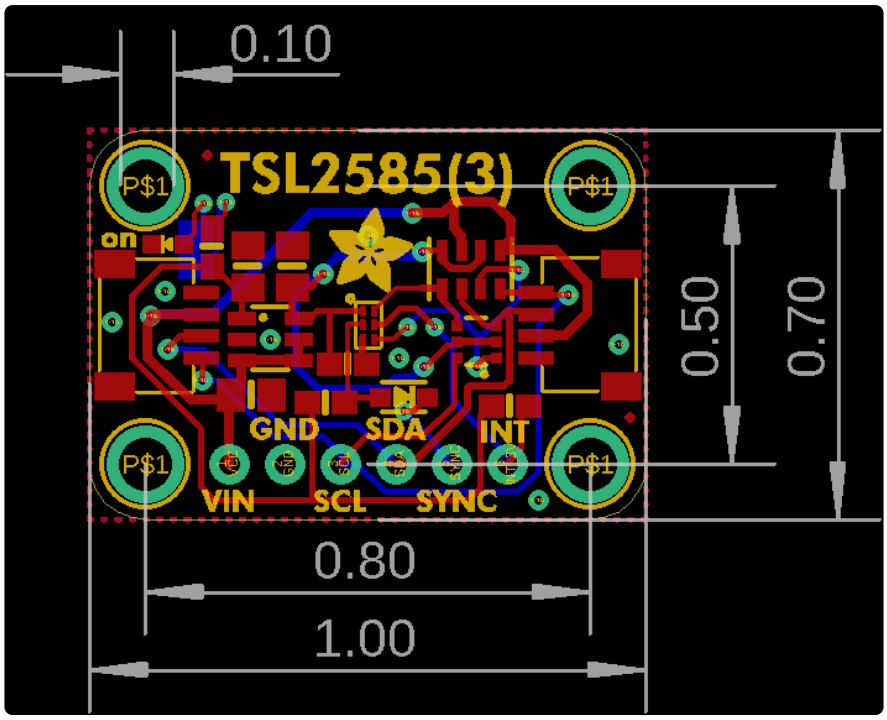
Downloads

Files

- TSL2585 Datasheet (http://adafru.it/652465242585001043500)
- EagleCAD PCB Files on GitHub ()
- Fritzing object in the Adafruit Fritzing Library ()

Schematic and Fab Print




0.10
TSL2585(3)
P$1
P$1
P$1
P$1
GND
SDA
INT
VIN
SCL
SYNC
0.50
0.70
0.80
1.00